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Short Communication

The solid reaction of lithium hydride and lithium hydroxide in lithium hydride pellet under normal condition and the application of CO_2 for long-time storage

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ABSTRACT

The solid reaction of lithium hydride and lithium hydroxide in lithium hydride pellet is first directly observed at normal ambient condition after 4 years storage. Layer-by-layer XPS results show that LiOH hydrolysis layer quickly forms on the LiH surface, LiOH reacts with bulk LiH to form Li_2O and H_2. Due to the great H_2 outgassing after long time conservation, bubbling problem actually exists during the practical long-time preserving period. Using CO_2 as pre-treatment gas during the LiH sample preparation process is a feasible way to prevent LiH sample from bubbling after long storage period. Moreover, Li_2CO_3 shows possible delay effect on the solid reaction of LiOH with LiH in this work, thus preserving LiH sample in CO_2 containing atmosphere might also be an prospective way to protect the sample from further corrosion.

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1. Introduction

Lithium hydride (LiH), with the combination of high hydrogen density and low weight density, is an attractive material for neutron shield, nuclear reactor fuel, hydrogen storage, etc. [1–6]. LiH has a high affinity for water, the reaction of LiH with H_2O is highly exothermic and extremely easy to form a surface hydrolysis layer of lithium hydroxide (LiOH) and release H_2 even exposure to trace quantities of moisture (Eq. (1)) [7–11].

$$LiH + H_2O \rightarrow LiOH + H_2(g) \quad (1)$$

Moreover, due to the instability of LiOH [12–14], accelerating experiments have shown that more H_2 generates through irradiation or temperature programmed reaction of LiH and LiOH hydrolysis [13,15,16], forming vesicles with the size of angstroms to microns [17–20]. And the H_2 evolution mechanisms vary in these reports [8,12,13,21–24].

In practical applications, LiH materials are always covered with protective films (usually water-barrier polymer) and stored under normal ambient condition for long time preserving. As the moisture is insulated while the LiOH hydrolysis layer is inevitable during current processing procedures, it is reasonable to speculate that the H_2 outgassing basing on LiH and LiOH contribute nearly the entire storage period and might cause bubbling problem when the inner H_2 pressure increased high enough, which even lead to the breach of containment and further corrosion after exposure to air. However, till now, nearly no such work has been reported under normal ambient condition, because the solid reaction of LiOH and LiH is too slow to be observed at room temperature and it is very difficult to control the reaction environment [9,14,21]. Thus, it is greatly necessary to study the real reaction of LiH sample under normal ambient condition for a long preserving period. Additionally, it is also urgently needed and challenging to effectively solve the possible bubbling problem.

In this work, we have prepared LiH sample covered by parylene C polymer film with negligible water penetrating through the film, and preserved the sample at room temperature for 4 years. Agreeing well with our previous speculation, bubbling problem actually occurres during the practical long-time preserving period, where big bubbles are observed and the polymer film separates from LiH substrate (Fig. S1). About 95% of the gas in the bubbles is H_2, and Li_2O is also detected on the bubbled LiH surface after 4 years storage (Fig. S2), indicating the solid reaction between LiOH and LiH. Layer-by-layer XPS analysis is utilized to get the chemical information of the hydrolysis layer of bubbled LiH sample by ionic beam bombardment with different time. Additionally, to solve the bubbling problem, new samples were prepared after treated with dry air containing about ~0.5 % CO_2 for 1 day. After storing for

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another 4 years, the LiH samples were free from the bubbling problem (Fig. S3). More interestingly, Li_2CO_3-rich layer was detected at the top of the hydrolysis layer by XPS analysis, showing possible inhibition effect of Li_2CO_3 on the solid reaction of LiOH with LiH. Thus, this after treatment method with dry CO_2-rich air exhibits great potential for long-life storage in industry.

2. Materials and methods

2.1. Samples preparation

LiH particles were pressed together and sintered into a monolithic structure at elevated temperatures, the cylindrical compact of LiH was machined into a size of 4 cm diameter × 5 mm thickness to obtain the LiH sample used in the following experiment. Then, the as-prepared LiH sample was polished with 1000 mesh SiC paper in a dry glove, and then transferred into a vacuum cavity and coated with a parylene C polymer film of 100 μm thickness using chemical vapor deposition method for 24 h. The water permeability of this polymer film was less than 1×10^{-11} g cm cm^{-2} s^{-1} Pa^{-1} as tested. Due to the high chemical activity of LiH, a small amount of surface LiH might be hydrolyzed into LiOH during the whole sample preparation process, resulting in the formation of LiOH hydrolysis layer on the LiH surface finally. Then the LiH samples with LiOH hydrolysis layer were stored in a glove box filled with dry Argon (relative humidity ≤1%) at room temperature for 4 years. Additionally, to solve the bubbling problem, new samples were prepared after treated with dry air containing about ~0.5 % CO_2 for 1 day.

2.2. Characterization

X-ray photoelectron spectrometer (XPS, Thermo-Fisher ESCALAB250) was employed to analyze the chemical component change after the isolated dry storage of the LiH samples. The vacuum in the XPS analysis chamber was better than 1.5×10^{-7} Pa. The Mg K_α line (1253.6 eV) was used for excitation of photoelectrons with 15 kV emission voltage. The photo beam spot was about 500 μm and the power was 300 W. The spectral resolution of the energy analyzer was estimated to be 0.6 eV. The LiH samples were transferred from glove box to XPS vacuum system after removing the surface protecting polymer. We used Argon ion beam to sputter and to analyze the depth chemical component information on the surface of the original and the stored LiH samples. Observed spectra position was affected due to the charge pile-up on the electrically insulating LiH. Therefore, the peak energy was corrected by the standard C1s peak.

3. Results and discussion

3.1. XPS analysis of the bubbled LiH samples

C 1s, O 1s and Li 1s spectra of the bubbled one at different sputtering time were shown in Fig. 1, and the atomic percentages of components within each region as a function of sputtering time were shown in Fig. S4.

3.1.1. C 1s region

Two different components were observed for the spectra (Fig. 1a), and the peak at 290.0 eV was assigned to Li_2CO_3 [25,26], which came from the chemical reaction of LiOH with CO_2 in air during the LiH sample preparation process. The decrease of Li_2CO_3 with increased sputtering time might be explained by the difficulty of CO_2 diffusing into the depths of LiOH hydrolysis layer within limited sample prepareation time (Fig. S4a).

3.1.2. O 1s region

As O 1s binding energies of LiOH (531.4 eV) [27,28] and Li_2CO_3 (532 eV) [25,26] were very close, the higher binding energy (around 531.5 eV) component corresponds to a mixture of LiOH and Li_2CO_3, and the lower binding energy (528.5 eV) [28] was ascribed to Li_2O (Fig. 1b); moreover, when sputtering time increased, the O 1s peak around 531.5 eV reduced while the peak of Li_2O grew. The decrease in LiOH and increase in Li_2O suggested the unceasing consumption of LiOH and formation Li_2O during 4 years storage at room temperature (Fig. S4b).

3.1.3. Li 1s region

A photoemission peak around 54.5 eV could be observed at the very beginning (Fig. 1c), which was ascribed to a mixture of Li_2CO_3 (55.3 eV) [25], LiOH (54.5 eV) [26] and Li_2O (53.6 eV) [29] on the surface. Upon sputtering to 11 min, the Li 1s signal was found to considerably broaden and shifted to the peak of Li_2O at 95 min, in agreement with the O 1s spectra. The simultaneous reduction of LiOH and LiH as well as increase in Li_2O consisted well with the chemical reaction of LiOH with LiH (Fig. S4c and Fig. S4d).

Basing on atomic percentage of C, O, Li and H (Fig. S4), the relative mol percentages of LiOH, Li_2CO_3, Li_2O and LiH at different depth of the bubbled LiH surface were also calculated and given in Fig. 2 (see detailed calculation method in the last part of Supporting Information). At the very beginning, Li_2O could hardly be detected on the original sample surface (Fig. S2); after 4 years storage, owing to the slow solid reaction between LiOH and LiH, at 11 min, the percentage of LiH and LiOH decreased to 39.16% and 34.43% respectively while Li_2O remarkably rose to 13.35%, the percentage of Li_2CO_3 kept almost the same. Additionally, as more H atomic existed in the inner layer of the LiH substrate, the percentage of Li_2O rose from 10.07% to 41.54% with increased sputtering time, namely with increased depths.

3.2. The solid reaction between LiOH and LiH at normal condition

The reaction of LiOH with LiH was a solid diffusion reaction that the positive H atom in LiOH combined with the negative H in LiH to form Li_2O and H_2 as Eq. (2).

$$LiOH + LiH \rightarrow Li_2O + H_{2(g)} \quad (2)$$

We would like to point out that this is the first report on direct observation of the Li_2O generation from the solid reaction between LiOH and LiH at normal room temperature condition, and LiOH would completely change into Li_2O and release more H_2 if it was stored even longer time. Additionally, according to the above equation, the more H_2 produce, the more Li_2O should formed. In Fig. 2, the bubbled sample showed remarkable high percentage of Li_2O, suggesting the rationality of our explaination.

3.3. The application of CO_2 for reliable long-time storage

Li_2CO_3 is thermostable and will not react with LiH once formed on the sample surface [30]; Inspired by this, if CO_2 is plenitudinous and the sample exposure time is long enough, LiOH hydrolysis layer might fully react with CO_2 to form compact Li_2CO_3 layer on the sample surface, and the reaction of LiOH with LiH can be effectively reduced, forming less Li_2O and H_2. Thus, it is reasonable to speculate that CO_2 can prevent LiH sample from bubbling after long storage period.

Our following studies lended support to the hypothesis above. The LiH sample preparation process was changed into CO_2 protective condition and the as-prepared sample was stored at room temperature for new 4 years. After such long preserving, no bubble was observed on LiH sample (Fig. S3). To know the detailed chem-

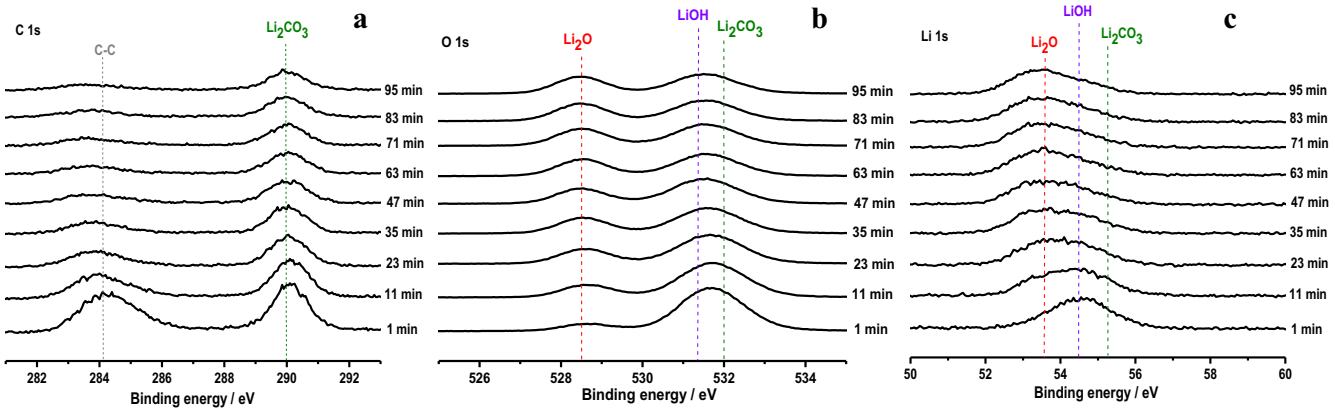


Fig. 1. Sputtering time dependent XPS spectra of the bubbled LiH sample in (a) C 1s, (b) O 1s and (c) Li 1s regions.

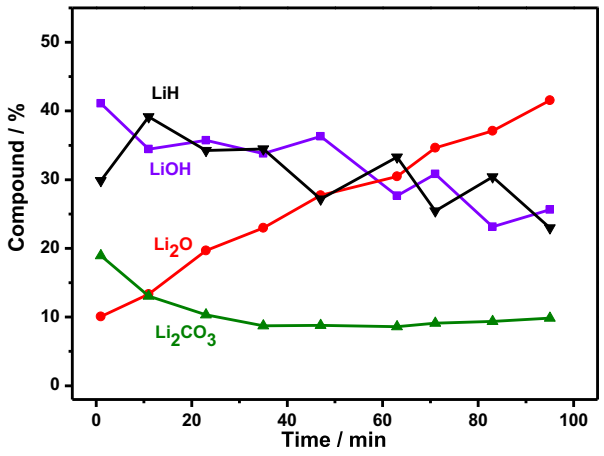


Fig. 2. Mol percentage of LiOH, Li_2CO_3, Li_2O and LiH in bubbled sample obtained from quantitative component analysis of atomic percentage of C, O, Li and H.

ical component information in different depth of the fine LiH surface, C 1s, O 1s and Li 1s spectra of the fine LiH sample at different sputtering time were shown in Fig. 3, and the atomic percentages of components within each region as a function of sputtering time were shown in Fig. S5.

3.3.1. C 1s region

Two different components were observed for the spectra (Fig. 3a). The peak at 284.4 eV was attributed to the hydrocarbon from protecting polymer film [25], and the other peak at 290.0 eV was assigned to Li_2CO_3 [26], which came from the chemical reaction of LiOH hydrolysis layer with ambient CO_2 during the LiH sample prepareation process. The decrease of both hydrocarbon and Li_2CO_3 with increased sputtering time might be explained by the difficulty of CO_2 diffusing into the depths of hydrolysis layer within limited sample prepareation time (Fig. S5a).

3.3.2. O 1s region

As O 1s binding energies of LiOH (531.4 eV) [27,28] and Li_2CO_3 (532 eV) [25,26] were very close, the higher binding energy (around 531.5 eV) component corresponds to a mixture of LiOH and Li_2CO_3 (Fig. 3b); however, when sputtering time increased, the broad O 1s signal reduced while the lower binding energy of Li_2O (528.5 eV) [28] appeared. The decrease in LiOH and increase in Li_2O suggested the unceasing consumption of LiOH and formation Li_2O during 4 years storage at room temperature (Fig. S5b).

3.3.3. Li 1s region

A peak centered at 55.3 eV could be observed at the very beginning (Fig. 3c), which was ascribed to Li_2CO_3 on the surface [25]. Upon sputtering to 35 min, the Li 1s signal was found to considerably broaden and included mixed peaks of Li_2O (53.6 eV) [29], Li_2CO_3 (55.3 eV) and LiOH (54.5 eV) [26]at 63 min; however, the Li 1s signal of Li_2CO_3 always existed during the successive sputtering time and its percentage was always higher than that in bubbled sample at different depths, indicating the more denser Li_2CO_3 layer in fine sample (Fig. 3 c and Fig. S5c). Moreover, the percentage of LiH, LiOH and Li_2O became stable after 59 min (Fig. S5d), which

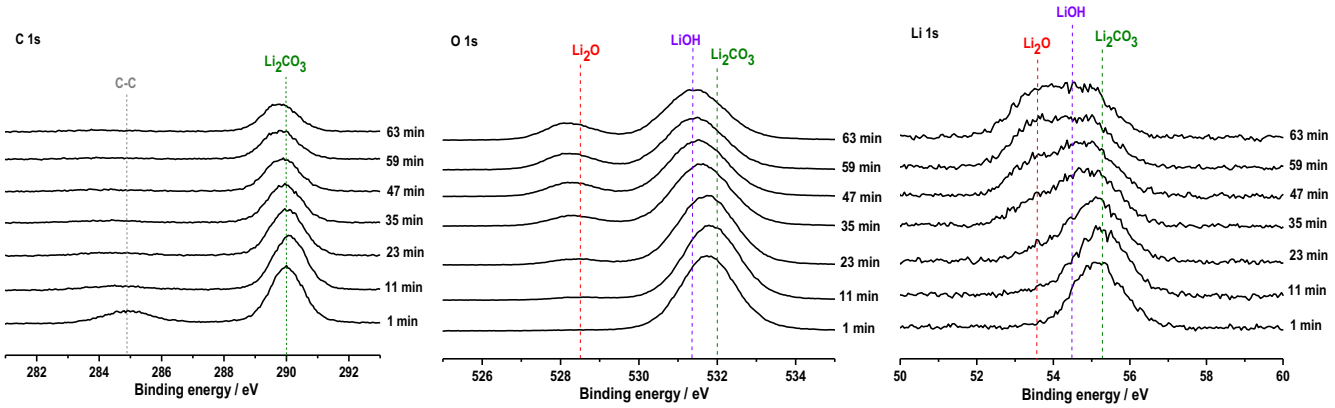


Fig. 3. Sputtering time dependent XPS spectra of the fine LiH sample in (a) C 1s, (b) O 1s and (c) Li 1s regions.

might suggest the achievement of solid reaction balance between LiH and LiOH.

The relative mol percentages of LiOH, Li_2CO_3, Li_2O and LiH at different depths of the fine LiH surface were also calculated and given in Fig. S6, nearly no LiOH or Li_2O existed on the fine sample surface while the percentage of Li_2CO_3 was extremely high (~65.20%) at 1 min, which was in good agreement with our previous hypothesis of the reaction between LiOH and CO_2 in LiH sample preparation process. More interestingly, previous XPS result on bubbled sample (Fig. 2) indicated that the percentage of generated Li_2O had a inverse relationship with that of Li_2CO_3; additionally, compared with bubbled sample, Li_2CO_3 was more denser while Li_2O was much less in different depth of the fine sample, suggesting the possible delay effect of Li_2CO_3 on the solid reaction of LiOH with LiH.

4. Conclusions

LiOH could react with LiH substrate at normal room temperature, which was a solid diffusion reaction that the positive H atom in LiOH combined with the negative H in LiH to form Li_2O and H_2. Due to the great H_2 outgassing after this reaction, bubbling problem actually existed during the practical long-time preserving period. Using CO_2 as protective gas during the LiH sample preparation process was a feasible way to prevent LiH sample from bubbling after long storage period. Moreover, Li_2CO_3 showed possible delay effect on the solid reaction of LiOH with LiH in this work, thus preserving LiH sample in CO_2 containing atmosphere might also be an prospective way to protect the sample from further corrosion.

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Appendix A. Supplementary material

Supplementary data associated with this article can be found, in the online version, at https://doi.org/10.1016/j.apsusc.2018.04.024.

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